

IST604 – Web Services & Distributed Computing

Chapter 2: Practice MCQ Questions

Instructions: Select the best answer for each question. Answers are provided at the end.

Section A: Fundamentals of Web Services

1. Which of the following best defines a web service?

- A) A website accessible through a browser
 - B) A standards-based, language-agnostic software entity that communicates over a network using standardized protocols
 - C) A local API library bundled within an operating system
 - D) A database accessible through SQL over the internet
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2. Which statement correctly distinguishes a web service from an API?

- A) APIs must operate over a network; web services do not
 - B) All APIs are web services, but not all web services are APIs
 - C) All web services are APIs, but not all APIs are web services
 - D) Web services and APIs are functionally identical
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3. Which of the following is NOT a key characteristic of a web service?

- A) Language-agnostic
 - B) Vendor-neutral
 - C) Requires a shared database
 - D) Standards-based
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4. A web service that enables a Java application on a Linux server to communicate with a .NET application on a Windows machine is demonstrating which core capability?

- A) Loose coupling
 - B) Service reuse
 - C) Interoperability
 - D) Transport neutrality
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Section B: SOA and Architecture

5. In a Service-Oriented Architecture (SOA), which role is responsible for registering and discovering web services?

- A) Service Provider
- B) Service Requester

- C) Service Broker / Registry
 - D) Service Consumer
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6. What is the correct relationship between SOA and web services?

- A) SOA is the technical implementation; web services are the blueprint
 - B) SOA is the architectural blueprint; web services are the technical implementation
 - C) SOA and web services are the same concept
 - D) Web services replace the need for SOA
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7. In the SOA model, which protocol/standard is typically used by the service broker for service registration?

- A) SOAP
 - B) REST
 - C) UDDI
 - D) WSDL
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8. Which communication model requires the client to wait for a response before continuing execution?

- A) Messaging-based asynchronous model
 - B) Document-based model
 - C) RPC-based synchronous model
 - D) Fire-and-forget model
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9. Which statement best describes the messaging-based communication model?

- A) The client sends parameters and immediately receives return values
 - B) The client sends an entire document and does not wait for a response
 - C) The client and server share the same memory space
 - D) Communication is always bidirectional and synchronous
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Section C: Web Services Standards

10. SOAP is best described as:

- A) An architectural style for building lightweight APIs
 - B) A W3C-accepted XML-based messaging protocol transported over HTTP, SMTP, or FTP
 - C) A registry for discovering and integrating web services
 - D) A language for describing the interface of a web service
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11. Which element of a SOAP message is **required**?

- A) Header
- B) Fault
- C) Attachment
- D) Body

12. What is the role of WSDL in a web services ecosystem?

- A) It transports messages between service providers and consumers
- B) It registers web services in a global directory
- C) It defines the formal interface contract describing a service's operations, data types, and endpoint location
- D) It handles authentication and security for SOAP messages

13. Which WSDL element is considered the most critical and groups the abstract operations of a service?

- A) `<types>`
- B) `<binding>`
- C) `<portType>` / `<interface>`
- D) `<service>`

14. UDDI querying returns which artifact describing the corresponding service interface?

- A) A SOAP Envelope
- B) A WSDL description
- C) A REST endpoint URL
- D) An OpenAPI specification

15. Which of the following correctly describes REST?

- A) A strict, XML-only messaging protocol
- B) A lightweight architectural style using standard HTTP methods and JSON/XML
- C) A registry for discovering RESTful services
- D) A security standard for encrypting web service messages

16. JSON and XML are both used for data interchange. Which statement is accurate?

- A) XML uses key-value pairs; JSON uses tag-based markup
- B) JSON is a W3C-recommended markup language; XML is a scripting language
- C) JSON is a lightweight, text-based alternative to XML, using key-value pairs and arrays
- D) XML is exclusively used by REST; JSON is exclusively used by SOAP

Section D: SOAP vs. REST

17. Which of the following is an advantage of REST over SOAP?

- A) Built-in message-level security via WS-Security
 - B) Support for multiple transport protocols (SMTP, JMS, TCP)
 - C) Faster performance due to lightweight JSON payloads and HTTP caching
 - D) Standardized Fault elements for detailed error reporting
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18. In which scenario would SOAP be the more appropriate choice over REST?

- A) Building a public-facing mobile app API
 - B) Designing a microservices architecture for a startup
 - C) Processing financial transactions requiring ACID compliance and end-to-end encryption
 - D) Creating a lightweight social media feed API
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19. A SOAP request for retrieving a user must be sent using which HTTP method, regardless of the operation type?

- A) GET
 - B) PUT
 - C) POST
 - D) DELETE
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20. Which of the following is a key design philosophy difference between SOAP and REST?

- A) SOAP is resource-based; REST is operation-based
 - B) SOAP is operation-based (e.g., `GetUserInfo`); REST is resource-based (e.g., `GET /users/123`)
 - C) Both use identical design philosophies but differ only in transport protocol
 - D) REST requires a formal WSDL contract; SOAP does not
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Section E: WSDL and Implementation

21. Which WSDL development approach starts with existing code and auto-generates the WSDL file?

- A) Top-Down (Contract-First)
 - B) Bottom-Up (Code-First)
 - C) Middle-Out
 - D) Schema-First
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22. In WSDL 2.0, the `<portType>` element was renamed to:

- A) `<endpoint>`
 - B) `<description>`
 - C) `<interface>`
 - D) `<binding>`
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23. Which WSDL operational pattern describes a service that sends a message to a client without expecting any response?

- A) Request-Response
 - B) One-way
 - C) Solicit-Response
 - D) Notification
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24. Which Java API is used specifically for building SOAP-based web services?

- A) JAX-RS
 - B) Spring Boot
 - C) JAX-WS
 - D) RESTeasy
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25. Which of the following tools would a developer use to test REST and SOAP web service endpoints?

- A) wsimport and WSDL2Java
 - B) Postman and SoapUI
 - C) wsgen and Java2WSDL
 - D) Jersey and RESTeasy
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Answer Key

Q	Answer	Q	Answer	Q	Answer
1	B	10	B	19	C
2	C	11	D	20	B
3	C	12	C	21	B
4	C	13	C	22	C
5	C	14	B	23	D
6	B	15	B	24	C
7	C	16	C	25	B
8	C	17	C		
9	B	18	C		